Soumodeep Karmakar

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Education

B.E. Electrical Engineering 2024 – Present
Jadavpur University, Kolkata *Kolkata, India*

- **Coursework:** Stanford CS336, CS229, CS231N, CS224N, Signals & Systems, DSP, Control Systems, Linear Algebra, Probability & Statistics, Numerical Methods & Optimization, Calculus & Multivariable Calculus, Fourier & Laplace Transforms

Experience

Research Contributor Jun 2026 – Present
CMATER Lab, Jadavpur University (under Prof. Ram Sarkar) *Kolkata, India*

- Working on **medical image segmentation**.

Software Team Member Aug 2025 – Mar 2026
Jadavpur University Mechatronics Club (JUMTC) *Kolkata, India*

- **IRC 2026:** Developed a real-time **computer vision pipeline** for a field rover, enabling automated mechanical tool detection across varied terrain and lighting conditions.
- Collaborated with a cross-functional team of 20+ students on system integration between the onboard vision module and ground station software.

Skills

AI & ML: Machine Learning, Deep Learning, LLMs, NLP, Computer Vision, Transformers, Image Segmentation, Object Detection, Signal Processing
Frameworks & Libraries: PyTorch, Triton, OpenCV, MediaPipe, scikit-learn, XGBoost, Librosa, NumPy
Languages: Python, C/C++, JavaScript
Developer Tools: Git, GitHub, Docker, Roboflow, Kaggle, VS Code, Modal

Projects

Kaizen | *PyTorch, Transformers, MoE* |

- Built a **decoder-only LLM from scratch** in PyTorch, implementing multi-head self-attention, **Multi-head Latent Attention (MLA)**, RoPE, RMSNorm, and SwiGLU feed-forward layers.
- Implemented a **Mixture-of-Experts (MoE)** layer with top-k routing and a custom byte-pair encoding (BPE) tokenizer built from scratch.
- Built the complete training pipeline: AdamW with weight decay, gradient clipping, cosine LR scheduling, and cross-entropy loss with vocabulary projection.

Hikari | *PyTorch, Triton, Distributed Training, Modal* |

- Implemented and benchmarked **three distributed data-parallel (DDP) strategies from scratch**: naive per-parameter all-reduce, batched flat-buffer all-reduce, and overlapped async-hook all-reduce during backward pass.
- Wrote a **FlashAttention forward-pass kernel** and a multi-GPU benchmarking harness on Modal to compare compute/communication overlap across strategies.

Flood Semantic Segmentation | *UNet++, ResNet-101, CBAM, PyTorch, SAR+Optical* |

- Built a multi-class flood segmentation system using **SAR + multispectral** satellite imagery; designed a custom **CBAM-based learnable adapter** to project N-band input into 3 channels for a frozen ResNet-101 encoder.
- Applied **Focal + Dice loss** with class weights and CutMix/GaussNoise augmentation to handle severe data imbalance (59 training images); implemented softmax-threshold post-processing to resolve class ambiguity.
- Achieved **Flood IoU: 0.1981** – **Top 4 Honorary Team, Top 7 Overall** at AISEHack 2026, IIIT Hyderabad (IBM Theme).

Nuclei Segmentation: U-Net Variants Study | *PyTorch, ResNet34, ViT, CBAM, Deformable Conv* |

- Benchmarked **9 architectures** for nucleus segmentation on TNBC histopathology (MoNuSeg): Attention U-Net, ResNet34 Attention U-Net, CBAM U-Net, SA U-Net, EViT-UNet, SE-UNet, DA-UNet, TransRes-UNet, and Base U-Net.
- Best result: **Dice 0.8652, IoU 0.7876, HD95 2.88 px** (Attention U-Net); pretrained ResNet34 encoder reached **Dice 0.8559 in only 10 epochs**, fastest convergence across all variants.

Achievements

AISEHack 2026 – Top 4 Honorary Team (Top 4 Overall) Apr 2026
IIIT Hyderabad | IBM Theme: Flood Detection

Medha-Manthan 2025 – 1st Prize Sep 2025
IEEE Kolkata Chapter | Assistive communication board for physically disabled people

Hackforge – 3rd Position Mar 2025
Srijan'25, CodeClub Jadavpur University